

17. Explain the purposes of assessment in education.
(A) [Ans: 1:06]
18. Describe the principles of good assessment practice.
(A) [Ans: 1:07]
19. What are the important stages in implementing an assessment programme? (C) [Ans: 1:08]
20. Explain the principles related to choosing / developing methods of assessment. (A) [Ans: 1:08:1 + 1:08:1:01]
21. State the principles related to collecting the assessment information and judging and scoring of student performance. (A) [Ans: 1:08:2 + 1:08:3]
22. State the principles related to summarizing, interpreting and reporting the assessment results.
(A) [Ans: 1:08:4 + 1:08:5]

UNIT – II

ASSESSMENT FOR LEARNING IN CLASSROOM

2:00 Introduction

Classroom assessment plays an important role in students' learning achievement. Classroom assessment is the process of gathering evidences of what students know, understand and able to do. Teachers attempt to evaluate students' learning in terms of the criteria formed, based on expected learning outcomes and level of performance. These criteria form the basis for evaluating students' learning progress and reporting the same to parents.

In this unit, Evaluation of students' performance based on the concept that education is reaching out to students what are to be learnt i.e. evaluation of students based on transmission-reception model, which is otherwise known as 'behaviourist model', its drawbacks, changes that are taking place in the assessment practices important among them being (i) Use of constructivist approach in the assessment of students i.e. giving more importance for how students learn than for what are learned (ii) Use of continuous and comprehensive evaluation along with

using grade symbols in the place of marks to denote learning achievement (iii) Adopting different techniques in evaluating multiple aspects of student development, different kinds of assessment (practice based, evidence based, performance based and examination based). Some important practices of assessment of student learning (dialogue, feed back through marking, peer and self-assessment) are to be discussed in detail.

2:01 Behaviourist View of Learning

According to behaviourism, 'Learning is nothing but expressing appropriate behaviour in response to the information received from the environment'. When we apply the behaviourist model to the classroom teaching-learning process, it means students receive knowledge based information or learning experiences from the teacher, the outcome of which is getting expressed in the form of student behaviour. According to this view point, in the teaching learning process, the teacher is very active and provides appropriate learning experience to students while students remain passive, receive the learning experiences provided by the teacher and get them registered in their memory, as a result of which students' responses emerge.

Learning experiences received by students find predominant place in the behavioural model of

learning. In this model, there is no place for higher order cognitive abilities like understanding what are being learnt, making use of the knowledge already one has acquired and getting adopted accordingly etc. Putting it in brief, the learning model advocated by behaviourists, treats learning as a mechanical process of '**Transmission-Reception**' in which the teacher provides appropriate learning experiences while the students receive and get them expressed in their behaviour. Hence, the knowledge acquired by students will remain only **informational** and not become **functional**, developing the ability to apply it according to the needs and requirements.

2:01:1 Student Evaluation in Transmission-Reception Model Based on Behaviourist Theory

In the transmission-reception model of learning which is based on behaviourist theory, what students have learned are only assessed and no importance is given for, how they have learned i.e., in the 'transmission-reception' model, assessment of students' learning is done by employing '**Product assessment approach**' and not by '**Process assessment approach**'. In other words, informational knowledge acquired as a result of learning are assessed through oral or written form of tests; but the way how learning took place, the skills abilities,

attitudes and values are developed as a result of learning etc. are not assessed. Stated briefly, learning assessment done based on the 'transmission-reception' model, evaluates only the informational knowledge of students and hence it is narrow in nature as it does not evaluate the learning outcomes comprehensively. Since very long, in our classroom assessment only the 'product approach' is in vogue.

2:01:1:01 Drawbacks of Assessment Based on Transmission-Reception Model

Using 'Product assessment approach' based on the 'Transmission-Reception' model has the following drawbacks in evaluating students learning:

1. It is narrow in nature, as it does not assess comprehensively the learning outcomes but assesses only the informational knowledge.
2. In the product assessment approach, it is very difficult to assess whether the students have really understood what they have learned, have they developed the ability to apply them in the practical life etc.
3. The 'product assessment approach' does not attempt to assess the proficiency achieved in pertinent skills, development of desirable attitudes and values .

4. This approach gives emphasis for summative assessment and not for formative assessment that tries to continuously improve student learning by removing the deficiencies found in it.
5. The product approach of assessing student learning encourages students to learn by heart and retain what are learned in their memory to be retrieved as and when required i.e., this approach of assessing student learning mainly attempts to evaluate students' ability to memorise and retain rather than their understanding.
6. This approach does not encourage students to develop higher order cognitive abilities like critical thinking, problem-solving, creative thinking etc.
7. The 'product approach of assessing students' learning achievement, mainly employs oral and written forms of tests. In this assessment of students' learning achievement except the informational knowledge acquired no other elements of personality development are evaluated
8. The product approach of assessment does not drive pupils for further learning.

2:02 Changing assessment practices

Till recently, much emphasis was given only for assessing informational knowledge acquired by students as a result of their learning. However, the main drawbacks of this approach like its narrowness, inability to improve student learning by removing the deficiencies found and driving students for further learning were realised by educationists. As a result, at present the practice of learning assessment has started changing as broad based, assessing all aspects of student learning like knowledge acquired, development of skills, ability, desirable attitudes and values, considering assessment as an integral part of the teaching- learning process, improving student learning through continuous feedback provided etc. The following are the major changes that have taken place in the assessment of student learning:

- i) Giving importance to the use of constructivist approach in assessment i.e. giving more emphasis for students to be active and participate directly in all learning activities, process every information they receive on the basis of their personal experiences and assimilate it as new knowledge by fitting it with their existing cognitive structure.
- ii) Use of continuous and comprehensive evaluation

- iii) Adopting different types of assessment
- iv) Use of grades instead of marks in the assessment of students' learning achievement.

Let us discuss each one of these in detail, in the pages to follow.

2:02:1 Use of Constructivist Approach in Assessment

2:02:1:01 Meaning of Knowledge Construction / Constructivism

Constructivism, as a perspective in education, explains how knowledge is constructed in the human being when information received comes into contact with his / her existing knowledge that had been developed by personal experiences. The learner processes every information he receives, on the basis of his personal experiences and assimilates it as new knowledge by fitting it with his existing cognitive structure, as a result of which the cognitive structure gets modified. Putting it in other words, learning is nothing but the process of constructing one's knowledge. Knowledge is not viewed as some sort of a true copy of the features of the world outside but the process of construction of the individual. The learner, gets information related to the objective reality of the environment around him, examines them in the

light of past and present personal experiences so as to make sense out of them and finally register them in the mind as subjective representations. i.e. **Objective reality of the world outside is processed and internalised as subjective knowledge of an individual.**

A constructivist classroom environment will facilitate students to critically examine any new information they receive by raising appropriate questions regarding that and trying to get answers for the same and allow for collaborative learning to find solutions for problems. The teacher will not be a person pouring the subject contents in the classroom. Importance will not be given for advancing correct answers to problems, expecting identical answers or explanations from students as the correct response for the question posed by the teacher etc.

Summing up, it could be stated that the following three constitute the principles of constructivism

- i) learning is an active process
- ii) the learner has prior knowledge on the basis which he/she examines any new information received to make sense out of it i.e. tries to understand the new information by associating it with existing previous knowledge

- iii) the learner takes responsibility for his/her learning.

2:02:1:02 Constructivism and Assessment

Constructivists believe that assessment is a tool to be used for improving both the student's learning and the teacher's understanding of his students. Those who adopt constructivist approach consider assessment as a formative process rather than a summative act. Its purpose is to improve the quality of student learning, not to provide evidence for evaluating or grading students. Assessment is an ongoing process along with learning.

For example, while starting a new lesson or topic, the teacher initiates by raising questions related with it. By asking questions, the teacher besides motivating students to learn the new topic, tries to test the previous knowledge of students regarding the subject content to be taught. During the second stage of classroom teaching viz. presentation of the subject content, the teacher besides making students understand every concept taught by raising suitable questions that relate it with students' previous knowledge, also ask them to cite examples from practical life wherein it finds use. Based on the responses of students, the teacher provides feedback which helps them to enhance their understanding further. In the terminal stage of classroom teaching, teacher evaluates student

learning by asking questions. Thus, at every stage of the classroom instructional process, student learning is assessed so as to improve and enhance it i.e. student learning and assessment go hand in hand, throughout the instructional process. Assessment of student-learning, under-taken at the terminal stage of the instructional process also serves as the basis for the next phase of students' learning. Hence it is no exaggeration to state that assessment of student learning forms a part and parcel of the instructional process that conforms to the constructivist approach. The primary aim of the assessment of student learning is to improve and enhance it rather than providing evidence for grading students according to the levels of their learning achievement.

2:02:1:03 Tools and Methods Used in the Constructivist Assessment

In the assessment of student learning that follows the constructivist approach, the teacher uses only those methods which allow for engaging students in verbal interactions like discussion, debate, story telling, dramatization etc. and judging the level of learning achievement by observing their performance in the learning activities. Besides observation, the teacher could also use tools such as 'student portfolio', 'investigatory projects', 'check lists', 'written' and

performance tests of achievement for assessing students' learning proficiency.

2:02:1:04 Role of Teacher in Constructivist Assessment

The role of teacher in the constructivist classroom environment, which provides for adequate students' initiatives, is as follows :

- i) Teacher will often puts open-ended questions in the class and patiently waits for getting student's responses.
- ii) Teacher will give more importance for students' higher order thinking and logical reasoning.
- iii) Teacher will ensure adequate opportunities for students to interact with him as well as among themselves.
- iv) Teacher will encourage students to discuss and mutually share their learning experiences.
- v) Teacher will encourage students to learn by investigating and critical thinking.
- vi) Teacher will give more importance for student learning, based on problem-solving.

- vii) Instructional techniques such as 'group interacting sessions', 'cooperative learning', 'collaborative learning', 'group project work', 'solving jig saw puzzles' etc. will be increasingly used by the teacher.

The following are the four phases in teacher's planning and implementation of students' cooperative learning in the classroom :

- i) Making decisions before the lesson begins.
- ii) Setting the learning activities for the lesson.
- iii) Monitoring students' activities while they work in groups.
- iv) Assessing the process and product of group work.

Stated briefly, the three important instructional techniques employed by the teacher in the classroom environment that helps students to construct their own knowledge are :

1. Stimulating students' inquiry related to every concept taught in the class.
2. Providing for multiple interpretations which help multiple intelligences get expressed.
3. Encouraging students' group work and use of peers as resources.

2:02:2 Use of Continuous and Comprehensive Assessment

2:02:2:01 Concept of Continuous and Comprehensive Evaluation - CCE

'Continuous and Comprehensive Evaluation' (CCE) refers to the practice of evaluating students' skills and proficiency that covers all aspects of development, periodically throughout the course and maintaining a record of the cumulative progress achieved.

The two adjectives of evaluation found in CCE viz. 'Continuous' and 'Comprehensive' are very important. In this scheme the term 'continuous' is meant to emphasize that evaluation of identified aspects of students' growth and development is a continuous process rather than an one time event, built into the total teaching - learning process and spread over the entire span of academic session. It means regularity of assessment, frequency of unit testing, diagnosis of learning gaps, use of corrective measures, retesting and also feedback of evidence to teachers and students for their self-evaluation. Putting in brief, it involves a minimum of two formative evaluations during each term / semester of a course and a summative evaluation at the end of each term / semester.

The second term 'comprehensive' means evaluating all aspects of students' growth and development i.e. growth and development in all the three domains – (i) cognitive (ii) affective and (iii) conative. Thus comprehensive evaluation is a holistic evaluation which includes both scholastic and co-scholastic proficiency achieved by students.

Since abilities, attitudes and aptitudes can manifest themselves in forms other than the written word, the term 'comprehensive' in CCE refers to application of a variety of tools and techniques (both testing and non-testing) and aims at assessing a learner's development in areas of learning like knowledge, understanding, applying, evaluating and creating.

2:02:2:02 Aims of CCE

The following are the important aims of CCE scheme.

- i) To make evaluation of students an integral part of teaching - learning process i.e. by assessing student - learning at regular intervals throughout the period of instruction and providing feedback to teachers and students to improve both teaching and learning.
- ii) To lay emphasis on thought process and de-emphasize memorization in learning.

- iii) To eliminate subjectivity or subjective bias in evaluating students' attainments.
- iv) To keep students continually motivated to learn throughout the academic year.
- v) To encourage students to participate in co-scholastic activities besides achieving scholastic proficiency.
- vi) To help the teacher take corrective measures based on the feedback received from the assessments of student learning.
- vii) To assess students' multifaceted development and encourage to improve further.
- viii) To make the process of teaching and learning a learner-centred activity.

Stated briefly CCE aims to use evaluation for improvement of students' achievement and teaching - learning strategies on the basis of regular diagnosis followed by remedial instruction. Further CCE aim to evaluate all aspects of students' development.

2:02:2:03 Characteristics of CCE

The characteristics of school-based CCE are the following :

- i) CCE involves the process of assessing student -

learning continuously throughout the academic year at regular intervals. It evaluates all aspects of learners' progress i.e. both scholastic and co-scholastic aspects of progress.

- ii) The scholastic component of CCE conceptualizes evaluation of all academic subjects spread over the entire span of the learning period (i.e. term or semester).
- iii) It also includes physical education in its scope of assessment.
- iv) It uses a variety of tools and techniques like written tests, observation, rating scale, check list, quiz etc. to evaluate different aspects of learner's achievement.
- v) CCE is carried out through 'Formative Assessment' (FA) and 'Summative Assessment' (SA).
- vi) FA is criterion based, diagnostic and remedial; it offers feedback to students helping them know their deficiencies in their learning and take appropriate remedial measures; it also provides feedback to teachers to improve their teaching by orienting towards the felt needs of students.
- vii) Descriptive indicators are used to assess the achievement profile of the learners in FA.

- viii) SA involves regular and norm-based assessment of students' achievement in all academic subjects at the end of a term / semester.
- ix) It lays emphasize on thought process and de-emphasizes memorization in learning.
- x) Co-scholastic evaluation is the assessment of many informal and formal developmental areas such as life skills, attitude and values, wellness, service activities and work education.
- xi) By finding out the individual needs of learners and make them achieve, helps students to become efficient citizens.

2:02:2:04 Functions of CCE

Important functions of the CCE are the following :

- i) It helps the teacher organize effective teaching strategies.
- ii) It helps the teacher to assess students' progress continuously.
- iii) It serves to diagnose the deficiencies in student-learning and permits the teacher to ascertain, the individual learner's strengths, weaknesses and needs. It provides immediate feedback to the teacher, who then decide whether a particular unit or concept needs a discussion again in the whole

class or whether a few individuals are in need of remedial instruction.

- iv) The feedback it provides helps students to have a realistic self-assessment of how he / she studies. It can motivate students to develop good study habits, to correct errors and to direct their activities towards the achievement of desired goals. It helps a learner to determine the areas of instruction that require more focus and emphasis.
- v) CCE identifies areas of aptitude and interest of students. It also helps the teacher in identifying the required changes in attitudes and values of students.
- vi) It helps in making decisions for the future, regarding choice of subjects, courses and careers.
- vii) It provides information / reports on the progress of students in Scholastic and Co-scholastic areas and thus helps in predicting the future success of the learner.

2:02:2:05 Merits of CCE

Merits of the CCE are the following :

- i) Student performance is continuously evaluated at regular intervals and feedback is provided for improvement in the teaching - learning process.

- ii) As student-performance is continually evaluated at regular intervals using tests, it promotes in students the habit of studying regularly i.e. students' motivation in learning increases.
- iii) Teaching - learning activities are well developed.
- iv) Students' attainment and progress are reported periodically to their parents and their cooperation in improving the learning of their wards is secured.
- v) CCE helps in identifying deficiencies in students' learning and taking immediate remedial measures for the same. Remedial teaching too is undertaken for those who lay behind in learning significantly.
- vi) It is non-threatening and helps in reducing learner's stress by assessing periodically meaningful small portions of the curricular content.

2:02:2:06 Disadvantages of CCE

The following are the disadvantages of CCE.

- i) Though CCE was introduced to minimise the examination stress of students, in reality after the implementation of CCE, students' mental stress and anxiety have increased as they have to take on increased number of assessment tests, conducted now and then.

- ii) Evaluation procedures differ from school to school.
- iii) Teachers' assessments are mostly skewed as a large number of the students are given 'A' grade (75% to 89%) for their performance; of which most of them do not deserve to get that high grade. Such students, on passing out the 10th standard examinations, join the science stream in higher secondary schools and struggle in their learning. In the 12th standard public examinations, these students mostly fail.
- iv) In over crowded classes with more number of students, proper implementation of CCE is practically not possible.
- v) Implementation of CCE has resulted in the increase of teachers' work load. Teachers have to spend more time in preparing evaluation reports, maintaining assessment records etc. rather than concentrating on their teaching work. As a result of this, teachers find little time for the preparation of classroom teaching, which leads to the deterioration of the quality of education provided.

2:02:2:07 Role of Teachers in CCE

1. Teachers should not spend more number of days in conducting assessment tests, allowing for the

decrease in the quantum of time available for regular classroom teaching.

2. Findings of assessment tests regarding students' attainment should be entered in the proper registers, then and there, without any delay.
3. Teachers should be honest and fair in evaluating students' performance.
4. Teachers should never use assessment as a tool to discipline the students.
5. Teachers should provide proper feedback to students, based on the findings in the assessment report and take measures to improve student-learning. Teachers should inform the parents of students now and then, about the learning - progress of their wards and secure their cooperation in improving student performance further.

2:02:3 Adopting Different Techniques in Evaluating Multiple Aspects of Students' Development

For evaluating cognitive abilities of students, teachers use written tests, developed on the basis of 'Benjamin Bloom's Taxonomy of Instructional Objectives'. However using this approach, attainments

in other skills and abilities of students could not be evaluated. As an alternative to this approach, the practice of evaluating student - performance, not only by written form of tests but also by the project work carried out, seminars participated, assignments submitted, student portfolio etc. has started gaining popularity in recent times. Let us discuss these in detail in the sections to follow.

2:02:3:01 Projects

(A) Meaning and Definition of 'Project'

The term 'Project' has been defined by a number of educationists in their own way. According to **Kilpatrick** "A project is a whole hearted purposeful activity proceeding in a social environment to learn the truth". **Stevenson** states "Project is a problematic act carried to completion in its natural setting".

In short, project is a piece of work or activity that is completed over a period and intended to achieve a particular purpose.

John Dewey initially promoted the idea of 'learning by doing'. Educational research has advanced this idea of teaching and learning into a methodology known as '**project-based learning**'.

(B) Project-based Learning

Project-based learning (PBL) is a student-centered pedagogy that involves a dynamic classroom approach in which it is believed that students acquire a deeper knowledge through active exploration of real-world challenges and problems. Students learn about a topic in a subject by working for an extended period of time to investigate and respond to a complex question, challenge or problem. PBL contrasts with paper-based, rote memorization or teacher-led instruction that presents established facts or portrays a smooth path to knowledge by posing questions, problems or scenarios.

Stated briefly, project-based learning is a style of active learning which involves finding a solution to a problem through investigation.

The core idea of project-based learning is that real world problems capture students' interest and provoke serious thinking as the students acquire and apply new knowledge in a problem-solving context. The teacher plays the role of facilitator and collaborator working with students to frame worthwhile questions, structuring meaningful tasks and coaching both knowledge development and social skills. The teacher constantly monitors students' participation in

the project and carefully assesses what they have learned from the experience.

Projects vary greatly in the depth of the questions explored, the subject content and activities required for their solutions and guidance available from the teacher. Projects can guide the entire curriculum or simply consist of some hands-on activities. They may be multi-disciplinary or based on single subject. Some projects involve the whole class, while others are carried out in small groups or individually.

C) Important Features of Project-based Learning

- i) Collaborative inquiry and investigation to solve problems results in students learning by direct experiences; their problem-solving skill gets improved; they get opportunities to create or make something novel.
- ii) Critical thinking, inquiry to learn, problem-solving, collaboration and various forms of communication, often known as "21st Century Skills" are developed in PBL.
- iii) 1. Choosing and proposing the project 2. Planning 3. Execution 4. Evaluation and 5. Recording are the five stages involved in project-based learning.

- iv) It incorporates feedback and revision with opportunities for self, peer and teacher assessment of content, essential deeper learning skills, student made products, research and presentations.
- v) Social skills like cooperation, collaboration, respecting others' views, team-spirit etc. get developed / improved while undertaking group projects.
- vi) Teacher could easily assess how students learn.
- vii) Practical knowledge and multiple skills get developed in students.

D) Advantages of Project-based Learning

- i) As students learn to work in a group, social responsibilities get developed in them.
- ii) It develops good work habits and attitudes.
- iii) It develops in students the ability to apply subject contents to the practical life.
- iv) It promotes self-directed learning and life long learning.
- v) It provides opportunities for the teacher to move closely with students in a meaningful way.

E) Demerits and Limitations of Project-based Learning

The following are the demerits and limitations of project method:

- 1) It absorbs a lot of time.
- 2) It involves much more work on the part of the teacher.
- 3) The whole syllabus, especially for more advanced classes, cannot well be included in a collection of projects and it is difficult to finish the syllabus in the limited time available.
- 4) Text books and materials written on these lines are not available.
- 5) This method is more expensive because students have to bear expenses of field work, outdoor activities, purchase of materials and do experiments. Further a well equipped library and laboratory are also required.

F) Assessing Students' Learning Through Project Work

In assessing students' project work the following aspects are to be taken into account.

- i) Significance of the problem selected for investigation.

- ii) Planning the details of gathering necessary data for the project.
- iii) Completing the project within the stipulated time.
- iv) Quality, appropriateness and the language style of the project report submitted.
- v) Participation and involvement in group work, if it is a group project.
- vi) Content proficiency and improvement in social skills achieved by completing the project successfully.

G) Illustration for Conducting a Project

In the project method, students should take some projects. Depending upon the nature of the project, the class may function as small groups or as a whole group. Sometimes the project itself may be divided into small units. Every student or every small group will be given an aspect or part of the project and they may be expected to complete it successfully. After the completion of the project, the activities and reports of all the students or groups will be presented to the whole class in the form of a detailed report or publication. Students will explain their experiences, information collected, findings obtained and the recommendations proposed.

For example, if the mosquito menace is taken as the problem for the project, a small group of students will investigate the causes of mosquito breeding; another group will survey the drains and stagnated pools of water in the area studied; yet another group will collect information regarding the various methods and techniques available for destroying the mosquitos; another group will implement the selected technique of eradication of mosquito menace in the area under investigation and test its effectiveness.

In the end, each and every group will record their activities and findings and prepare a report, to be submitted to the class teacher. The teacher will compile the reports of all the groups and release it as a comprehensive project report.

H) Characteristics of a Good Project

The following are the characteristics of a good project:

- i) Project should allow the active participation of both learners and teacher
- ii) It should be useful and purposeful
- iii) It should have definite educational value
- iv) It should be practicable
- v) It should provide maximum number of activities

- vi) It should not be expensive
- vii) It should promote the investigative ability among the learners.

2:02:3:02 Seminars

A) Concept of Seminar

Seminar refers to a single person lecturing or presenting a paper on a given topic or many persons delivering lectures or presenting papers on various topics. Generally this method is used more often in higher education. At the university level, research scholars present their theses or dissertations explaining the aims and objectives of the study undertaken, its scope and significance, possible benefits and implications to follow, limitations of the study etc. Generally 10 to 15 persons take part in a seminar. An expert in the particular field (possibly a university professor or the research guide of the candidate) serves as the chair person of the seminar. The members of the audience attentively listen to the lecture(s) in the seminar, at the end of which they raise questions regarding their doubts and seek clarifications. They can express their views and opinions also with regard to the replies given by the speakers. The chair person too can provide his views. In a seminar, more than one can speak, of course on

different topics which need not necessarily be related. A discussion may take place between the speakers and the members of the audience. It is the duty of the chair person to regulate the proceedings of the seminar.

B) Employing Seminar Method of Teaching at the School Level

At the high and higher secondary school levels also, seminar could be used as a suitable method to teach many a topic. Mostly seminar method is used alone i.e. other teaching strategies cannot be used along with seminar. When the topics are easy, the teacher can ask the students to prepare an essay or a write-up of two to three pages on the topic and present it to the class. The topics may be assigned for all the students. The papers may be presented by students in the class one after the other. After the presentation of a paper, students other than the one who presented it, can ask questions and raise doubts. The paper presenter will answer these questions. The teacher will be the chairman of the seminar. The size of the class will be small containing a maximum of 20 or 25 students. The class teacher acts as a resource person too. In the seminar method of teaching, it is possible to achieve higher levels of cognitive objectives like analysis, synthesis and evaluation.

(C) Merits of Seminar Method of Teaching

- i) It develops the habit of extensive reading of books in the students.
- ii) It develops in the students the habit of collecting relevant information pertaining to the topic on hand from various sources.
- iii) It removes shyness from the students and develops in them the language ability to present appropriate ideas logically and that too fluently.
- iv) It develops in students the spirit of participation.
- v) It develops the ability to face their students and answer questions boldly.

D) Disadvantages of Seminar as a Method of Teaching

- a) It requires more time, more space and more personnel.
- b) Sometimes only a few students will participate in the seminars; most of the students are not able to participate in seminars as they could not make proper preparations. Even if they attempt to participate in seminars, their lecture/paper presented may not be of quality.

- c) This method could be used only at the higher education level as it requires enough mental maturity on the part of the students to independently collect the required information, analyze and arrange them logically to put them in the form of an essay.
- d) Unless the chairperson is skillful, the seminars easily degenerate into a question and answer session or a lecture.

(E) Evaluating Student's Participation in Seminar

The following aspects of student's participation in seminar should be taken into consideration while assessing the student performance.

1. Concepts that found place in the seminar paper presented, explanations and illustrations offered for the same, and the appropriateness of the concepts advanced.
2. Student's ability to gather information related to the topic.
3. Student's communication skills.
4. Use of information technology in the presentation of the seminar paper.
5. Ability to function efficiently in clarifying the doubts raised by the members of the audience.

6. Ability to raise appropriate questions and seek clarifications at the end of the seminar, if a student participates in it as a member of the audience.

2:02:3:03 Assignments

A) Meaning of 'Assignment'

The dictionary meaning of the term 'assignment' is allotment to some one a particular job or work. In this sense, the term assignment in the teaching - learning process refers to the learning activities allotted by the teacher to students which may be either drill and practice of knowledge already acquired or carrying out theoretical study or practical activities on individual or small group basis, aimed at the realization of stipulated instructional objectives.

(B) Procedure for Using Assignment as a Teaching Device

1. The syllabus prescribed for the whole year is to be divided topic-wise into suitable parts. After dividing the subject contents into many small parts called 'assignments', the teacher should decide which ones are to be taught by him in the class every week or month and those that are to be allotted as assignments to students.
2. After allotting assignments to students, they may be instructed to finish their assignments within the

specified period of time, of course allowing adequate time to complete the assignments.

3. Teacher should give relevant references and guide the students in the preparation and completion of assignments.
4. Initial assignments are to be simple and easy to complete; slowly the difficulty level of the assignments may be increased in steps. This will help to sustain the interest level of the students.
5. More number of assignments may be given on theoretical topics. When assignments are given in practical work, teacher should provide necessary suggestions and guidelines for handling the apparatus and performing the experiments.
6. No student should be allowed to proceed to the next assignment without completing the previous one.
7. Specified time should be assigned in the class for submitting the completed assignments and undertake work related to the completion of the assignment.
8. Students who have not finished the theory part of a certain topic should not be allowed to go ahead with practicals related to these theoretical concepts.

9. Students may be instructed to keep a written record of their assignments. This helps the teacher as well as students to evaluate the progress of the teaching-learning process.

C) Advantages of Providing Assignment

1. Keeping in mind the individual differences among the students, the teacher could allot assignments to each student according to his mental ability, capacity and interest. Students too could proceed at their own speed according to their abilities and complete the assignments given to them.
2. Giving assignments to students helps to develop desirable habits in them such as
 - i) sense of responsibility to finish the task undertaken
 - ii) Undertaking self-study and work with self confidence
 - iii) Self-dependency in action and thought
3. It provides students the necessary freedom in learning activities. Students may go to library or laboratory and gather the required information; they may get hints or guidance from teachers as well as their own classmates to complete their assignments. As the duration for the execution of

assignments depends upon the abilities of individual students, they can work without any stress or anxiety.

4. The rapport between the teacher and students increases.
5. As a written record is maintained regarding the assignments completed by each and every student, the teacher could know the abilities of students and their progress in learning. While completing the assignments students too could understand their respective strengths and weaknesses.
6. As students pay more attention on executing their assignments, the problem of student indiscipline gets reduced considerably.
7. Based on the principle of 'Learning by doing', assignments promote better student - learning.

D) Limitations and Disadvantages of Assignment as a Teaching Device

Assignment as a teaching device suffers from the following limitations.

i) Not suitable for all types of students :

Assignment as a teaching device is not suitable for slow learners and those who lack interest in

learning. It is suitable only for those who have average level of intelligence and more as well as those who evince keen interest in learning more and more things.

ii) Gives room for cheating

Sometimes students may not prepare their assignments; instead may resort to copying from others.

iii) Increases the work-load of the teacher

As the teacher has to undertake responsibilities like allotting every student assignments, providing the necessary guidance for students to execute and complete their assignments and evaluating the submitted assignments, besides his regular teaching work, it increases strain on the teacher.

iv) Proves Expensive

As the assignment topics along with the relevant references and bibliography are either to be cyclostyled or printed by the teacher for giving to students, it incurs additional expenses for the teacher / school management. Further, students too have to spend considerable amount of money and time in executing and completing the assignments.

v) Other Practical Difficulties

There are many practical difficulties in adopting assignment as a teaching device in the prevalent conditions in our country such as

- a) Crowded classes with large number of students
- b) Paucity of good text books and reference books
- c) Difficulty in covering the heavy syllabus
- d) Non-conformity of the present examination system with assignment system in teaching and evaluation.
- e) Paucity of teachers, oriented to use 'assignment' as a teaching device.

E) Suggestions for Improving 'Assignment' as a Teaching Device

1. Instead of using 'assignment' as an independent method of teaching, this could be used along with other teaching devices i.e. assignment should be integrated with the regular classroom teaching. The teacher after presenting the content in the class either through a lecture or demonstration, could use the assignment as supplement to fix learning in the minds of students.

2. Teachers should plan and prepare assignments well in advance on a variety of topics in the subject.
3. Assignments planned for students should be well graded and sequentially arranged, starting with simpler ones and ending with those with considerable difficulty.
4. Students should be permitted to execute and complete their assignments in the library, laboratory and at home.
5. Assignments may be provided either to help students in the revision and application work or preparing for advanced study.
6. Assignments can be synthesized with projects so as to make teaching useful for students in their day-to-day life.

F) Assessing Students' Learning Based on Assignments Submitted

Assignments given could be completed either in the class or at home and submitted to the teacher within the specified period of time. As it is difficult to finish an assignment within the class duration of 45 minutes, mostly homebound assignments are given.

Teacher distributes the assignment topics among students groupwise and asks them to gather appropriate information related to their respective topics, prepare the assignment in the form of an essay and submit it individually within the stipulated time limit. Teacher provides students the particulars of reference books, addresses of websites etc. related to the topics of assignment and guides them in preparing the assignment given to them.

The assignments submitted by each member of the group on their respective topics will be distributed to other members of the group, so that each one will evaluate the assignment of another in the group i.e. peer assignment is arranged for. In order to ensure objectivity in the evaluation of assignments, teacher prepares the guidelines for scoring each of the assignments given and provides a copy of the same to each member of the respective groups. This arrangement makes it possible to get the assignments of all the students evaluated in a short time. If any doubt arises in scoring assignments, clarifications could be obtained from the teacher. The classmate / peer who evaluated the assignment of a particular student, will provide him / her the feedback based upon the assessment done. This practice, besides reducing the work load of the teacher, provides the opportunity

for all students in the class to know the information expected for each of the assignment topics and discuss the same with their classmates.

2:02:4 Students' Learning Portfolios

2:02:4:01 Meaning of Student Learning Portfolio

As an evaluation tool for community - based programmes, we can think of a portfolio as a kind of documents and photo album that records the progress and activities of the programme and its participants and showcases them to interested parties both within and outside the programme.

Similarly in the educational setting also, to assess the learning progress and achievement of students, their various learning activities could be documented and made available to the parents, teachers and educationists, showcasing how learning took place in students. A true learning assessment requires comprehensive evaluation of students in which besides marks obtained in written tests, various learning activities undertaken and learning experiences acquired are also taken into consideration.

The core of portfolio assessment of student-learning is providing opportunities for students to demonstrate rather than tell about what they know and

can do i.e. **Collecting and assessing information or data regarding student-learning from various sources, through multiple methods and over multiple points of time is known as 'portfolio assessment of student-learning'**. Contents of student - portfolios (sometimes called "artifacts" or "evidence") can include drawings, photos, video or audio tapes, writing or other work samples, computer discs and copies of students' answer scripts for standardized or subject-specific tests. The contents of the portfolio submitted are how far true and reliable could be ascertained from sources like the teacher, parents, staff of the educational institution and other community members who very well know the participants i.e. student participation in learning could be ascertained from different sources. Besides this, students' self reflections of their experiences too will be taken into account in the assessment of student-learning.

Note : Camp record, SUPW record, Audio-Visuals record, micro-teaching record etc. submitted by the teacher-trainees in the B.Ed. examination could be considered as their learning portfolios.

2:02:4:02 Definitions of Student Learning Portfolio

According to the definition advanced by **Paulson**

and **Mayer**, "Student learning portfolio is a collection of student's work that exhibits the student's efforts, progress and achievements in one or more areas.

Grace defines student portfolio as "a record of student's process of learning. What the student has learned and he / she has gone about learning, how he / she thinks, questions, analyzes, synthesizes, produces, creates and how he / she interacts - intellectually, emotionally and socially with others".

According to **DeFina**, "Student learning portfolios are systematic, purposeful and meaningful collection of student's work in one or more subject areas".

Summing up we can say, student learning portfolio is a systematic and purposeful collection of evidences in a given subject of study which reflect the level of learning achievement exhibited by the student and the efforts put in over a period of time to achieve the same.

2:02:4:03 Assessing Student's Learning by Using Learning Portfolio

Assessing student - learning in this method involves the teacher to take the following steps :

- i) Teacher selects the various learning activities that could be undertaken by students in the particular subject content area taught.
- ii) Teacher lists the selected learning activities based on the objectives of assessment.
- iii) Teacher then determines the criteria for assessing every learning activity.
- iv) Teacher clarifies to the students about the various learning activities that could be undertaken in a given subject area, the objectives and criteria of assessment of each one of these activities.
- v) Over a period of time, the teacher analyses and evaluates the learning activities of each student, based on the criteria of assessment which have been already clarified to students and assigns marks or grade accordingly.

For example, the science teacher could evaluate, how well the students have learned the important concepts of the lesson 'Newton's Laws of motion' by using the learning portfolio of each student as follows :

1. Students stating Newton's laws of motion in their own words.

2. Citing illustrations for each one from the day to day life.
3. Developing an improvised aid for explaining each one of the three laws of motion.
4. Preparing an essay describing what will have happen if each of the three laws of motion is not functioning.
5. Solving the given numerical problems, using Newton's laws of motion.
6. Explaining Newton's laws of motion by drawing suitable diagrams.

In the same way, the language teacher could observe in the classroom the learning activities of each student to assess the proficiency attained in the following language skills.

- i) Ability to express the ideas
- ii) Ability to interpret the information / data obtained
- iii) Skill of presentation of one's ideas
- iv) Skill of comparing and contrasting
- v) Skill of participating in group discussion
- vi) Skill of functioning as a team

In addition to the above, audio discs containing the recorded speeches of the student, essays, seminar texts presented, project reports submitted, answer sheets of tests, poems composed etc. which may find place in the student portfolio could be evaluated to assess the language proficiency of each student in the class.

2:02:4:04 Designing the Portfolio Assessment of Student - Learning

Three main factors guide the design and development of a student learning portfolio. They are : (i) Purpose (ii) Assessment criteria (iii) Evidences for assessment. Let us see them in detail.

i) Purpose

The first step in designing the student portfolio is to clarify the purpose that the portfolio will serve. For example, before getting started it should be clearly stated whether the portfolio is going to be used for assessing student's learning progress, reporting to parents about the learning activities of their children, identifying the special needs of students or fixing the person for programme accountability or for all of these.

ii) Assessment Criteria

Once the purpose or goal of the portfolio is clear, decisions are made about what will be considered as

success or good performance (Criteria to judge) and what strategies are necessary to meet the goals. Keeping these in mind, items are then selected to include in the portfolio because they provide evidence for meeting criteria or making progress towards the goal.

iii) Collecting Evidences for Assessment

It is necessary to collect evidences for assessment from various sources of which the following are the important ones.

a) Artifacts Produced : These are the items produced in the normal course of classroom or programme activities. E.g. Essays written by students in a particular subject content area, improvised aids developed, audio and video discs prepared etc.

b) Reproductions : Documentation of the interviews or projects done outside the classroom.

c) Attestations : Statements and observations by teaching staff, librarian and other staff of the educational institution testifying the participation of the student in different learning activities.

d) Production : Items prepared especially for the portfolio such as participant reflections on their learning experiences which reveals the involvement of the student in learning.

2:02:4:05 Advantages of Using Portfolio Assessment of Student - Learning

Portfolio presents a wide perspective of the learning process for students, enabling them to get a continuous feed back about their learning. It also helps students to have self assessment of their learning and progress further. Portfolio assessment benefits all stake holders in education such as teachers, students and parents, some of which are as follows :

- i) Portfolio assessment helps to assess student learning over a period of time in multiple ways.
- ii) It helps to assess student's proficiency in a given area of subject content in a realistic situation than paper - pencil tests.
- iii) It allows students, teachers and parents to assess the strengths and weaknesses of students' learning.
- iv) **It helps students to demonstrate their learning rather than describe it.**
- v) It helps students to develop the necessary abilities to become independent and self directed learners.
- vi) It helps students to use creative ways to share what they have learned.
- vii) It improves the relationship among the students, teacher and parents, as it gives a detailed

information about student's progress in learning i.e. it documents the process of learning with substantiative descriptions or examples of what the student is doing and experiencing while learning the academic content.

2:02:4:06 Disadvantages of Portfolio Assessment of Student - Learning

1. It helps more to assess the quality of student - learning rather than evaluating quantitatively the learning proficiency achieved by students i.e. it is considered less reliable in measuring the learning achievement of students as compared to written tests.
2. It is time consuming for the teacher to organise different learning activities in a particular content area of the subject and evaluate how each student performs in each of these learning activities. If portfolio assessment has to be done, in addition to traditional testing and grading, then teacher's work load increases manyfold. Portfolio assessment of student learning is not practically possible in overcrowded classes with more number of students.
3. As portfolio assessment requires teachers to develop their own individualised criteria for evaluation of student - learning, it can be felt too difficult; for some, it may be quite an unfamiliar act.

4. If goals and criteria are not clear, then the portfolio will be just a miscellaneous collection of artifacts. Using such portfolios, it is not possible to assess students' learning progress and achievement.
5. Like any other qualitative data, data from portfolio assessments are difficult to analyse or aggregate, to decide the level of learning achievement of students.

2:02:5 Use of Grading in the Place of Marks in the Assessment of Student Performance

Grading in education refers to classification of students into a few graded categories according to their level of achievement in examinations in every subject of study. Grades can be assigned as letters [generally A to E (Some prefer to use nine point grading scale as A_1 , A_2 , B_1 , B_2 , C_1 , C_2 , D_1 , E_1 , E_2 to increase the degree of differentiation) or as a range 1 to 5 (Some use a point scale 1 to 9)].

Grading system was developed by **William Farish** and first implemented by the University of Cambridge in 1792. Different countries use different grading scales in the assessment of students' performance. In India, following the introduction of Continuous and Comprehensive Evaluation System (CCES), Central

Board of Secondary Education (CBSE) started using the grading system with nine point grading scale from the academic year 2010-11 for Stds. VI to X. Tamilnadu Government too is following the grading system from 2012-13 for Stds. I to VIII and from 2013-14 for Stds. IX and X.

2:02:5:01 Important Features of the Grading Scheme Followed by the Tamilnadu Government

1. In all students' certificates and school records of students' achievement, only grades and not marks will be mentioned, pertinent to the level of performance.
2. The practice of declaring 'pass' or 'fail' has been stopped.
3. The result of candidates regarding their achievement is declared in two categories - (i) Eligible for qualifying certificate and (ii) Advised to improve performance.
4. There will be no mentioning of a thing like 'overall marks'; only 'Grade Point Average' (GPA) is mentioned.

All candidates even if they (scoring less than 33%) in one or all subjects will now have five chances

to improve their performance without having to repeat a year.

5. Marks obtained by students are to be transformed into a nine point scale, and the pertinent grade corresponding to the level of achievement be marked.

The following table illustrates the nine point grade scale. The minimum level for qualifying in any subject is grade 'D' and above.

Marks Range	Grade	Grade Point	Verbal Discrimination of Grades
91-100	A ₁	10.0	Excellent
81-90	A ₂	9.0	Very Good
71-80	B ₁	8.0	Good
61-70	B ₂	7.0	Good
51-60	C ₁	6.0	Fair
41-50	C ₂	5.0	Fair
33-40	D	4.0	Adequate Performance
21-32	E ₁	No Grade Point	Inadequate Performance; needs improvement
0-20	E ₂	No Grade Point	Poor Performance; needs huge improvement

6. Those whose achievement level is below the grade 'D', in any one or in all the subjects, will have five chances to improve their level of performance.

7. Those who fail to obtain minimum Grade 'D' in any one or in all the subjects at the Secondary School Examinations shall be given the first chance to improve their level of performance in the examination to be held in about a month, in the same year. Those who succeed in improving their level of performance could study Std. XI, without wasting a year.
8. For calculating the percentage of marks obtained by a student in a subject from the Grade Point, the following formula is to be used.

Percentage of marks
in a subject = $\text{G.P in the subject} \times 9.5$

9. A student's Grade Point Average (GPA) is obtained by dividing the 'Cumulative Grade Points' (CGP) [i.e. Total of the Grade Points obtained in all the subjects] by the total number of subjects.
10. The overall percentage of marks of a student is calculated by the formula $9.5 \times \text{GPA}$.

2:02:5:02 Uses of Grading System

As achievement of students could not be assessed precisely like measuring the external features of objects, grading of student performance using grading system helps in the following ways :

- i) It reduces the mental stress of students as it eliminates the thinking that those who score 59% are less proficient than those having an achievement score of 60%.
- ii) Students having scores lying in a particular class interval are considered to be equal in their level of proficiency.
- iii) A student's grade does not get influenced by the achievement level of other students.
- iv) It avoids competition among students.
- v) Students who fail to achieve the minimum level of proficiency accepted as adequate, are given a chance to improve their performance and get admitted to the next higher class, without wasting an academic year.

2:03 Types of Assessment

In education, assessment is an integral part of the teaching - learning process. It requires for the teacher to make use of different evaluative methods and tests to assess students' knowledge, understanding, skills, attitudes developed as result of teaching. Finding the level of learning achievement of students by evaluating their responses is not the only aim of student assessment. Providing meaningful feedback and

practice based on assessment to improve student learning is also emphasized. Different forms of assessment method are involved in evaluating various aspects of student learning such as the development of cognitive abilities, physical and social skills, attitudes and values. Of these different types of assessment, the following four, are important.

- i) Practice-based Assessment
- ii) Evidence-based Assessment
- iii) Performance-based Assessment
- iv) Examination-based Assessment

2:03:1 Practice-based Assessment

Assessing student - learning in the practice setting is one of the very complex forms of evaluation. Practice-based assessment was initially used in professional courses like "Teacher Education", "Nurse Training", "Medical Education" etc. for evaluating how far students apply efficiently the knowledge and skills they have learned in the true professional setting. For example, the primary aim of the B.Ed. Course which prepares teachers for secondary education, is to develop teaching skills, classroom management skills, leadership quality, communication skills, ability to work as a team etc. in student-teachers so as to make them

good and efficient teachers. After imparting in teaching skills through micro teaching sessions, student - teachers (teacher trainees) will be made to observe the classroom teaching of the mentor (guide teacher in the school) for a few sessions and then they will be asked to practise regular classroom teaching in the presence of the mentor. At the end of each such practice-teaching session the mentor will provide feedback to the student-teacher to improve his / her teaching. Next day, using this feedback the student - teacher will practise classroom teaching in the presence of the mentor, on a different topic or content area of the subject which will be again observed and assessed by the mentor and feedback provided. This cycle of student-teacher practising classroom teaching – Mentor observing and assessing the teaching and providing feedback based on the assessment – again student - teacher practising classroom teaching will take place a number of times (about atleast 10 times in a school subject), at the end of which the mentor will award the student - teacher marks for his/her teaching competency.

The important feature of such practice-based assessment is providing for practice with feedback so as to improve the learning achievement.

In the classroom environment of the school to undertake practice-based assessment, after teaching a topic or content area, an achievement test is to be conducted, based on which feedback is given to each student regarding the shortcomings found in his/her learning and the ways and means of eliminating them, so as to improve learning. This cycle of, teaching-conducting achievement test - providing feedback based on assessment for improving learning - testing again and providing feedback, should run atleast three times. Finally, the best among the three assessment scores is taken as the learning achievement score of that student. 'Practice with feed back improves student - learning' is the basis for this type of assessment. The achievement test conducted may be in the form of an oral test, written test or performance test. Assessing and judging student-learning through a single test is not fair. Providing opportunity for students to improve their learning through practice with feedback and then assessing the best performance expressed is the correct approach in assessment and that is what practice-based assessment advocates. In this approach, apart from assessing students' knowledge and skills, the attitude towards rectifying mistakes based on feedback provided and the virtue of continually striving for improvement through persistent efforts are also encouraged. The apparent

truth is "any learner could be made to achieve his/her learning goal; more intelligent students may require less practice while those with less intelligence need more amount of practice".

2:03:2 Evidence-based Assessment

Evaluating student achievement of expected learning outcomes is referred to as evidence-based achievement. It involves determining the 'Expected Learning Outcomes' (ELOs) of teaching a content area of a subject, then arranging the teaching - learning process based on the ELOs selected, gathering evidences for the realization of ELOs during teaching and assessing them. This kind of assessment of student learning is essentially formative assessment in its nature. Based on the assessment of the evidences gathered, during the teaching-learning process, the teacher can incorporate necessary changes in it i.e. efforts could be taken to introduce more appropriate learning activities in teaching.

Evidence-based assessment was first introduced in the field of medicine and has been introduced later on to other fields including education.

After clarifying the instructional objectives before start teaching a particular content area of the subject, teachers can decide how they will gather evidence of

emergent learning, realising the instructional objectives. There is no single way to gather formative evidence because formative assessment is not a specific kind of test. For example, teachers can gather evidence through interaction with students, observing how students carry out their tasks and activities like participation in classroom discussions, oral tests, quiz, practical work etc. or analysis of their work products like poems composed, illustrative charts, diagrams and models prepared etc. Whatever may be the method used to collect evidence of student learning, it should be a systematic and well planned process, yielding stream of information tied to indicators of progress in student learning.

Briefly stated evidence-based assessment is not assessing students' learning achievement at the end of the teaching - learning period; it involves assessing student-learning during the period of instruction. Gathering of evidences for student-learning takes place as teaching is in progress. Teacher should ensure that the evidences gathered for assessment of student - learning are based on the instructional objectives set out before start teaching a particular area of the subject content. The learning evidences created by a student in an academic year may turn to be his / her 'Learning Portfolio'.

The following table illustrates the methods of gathering and assessing direct and indirect evidences of student learning.

S. No.	Assessing the direct evidences for student-learning	Assessing the Indirect evidences for student-learning
1.	Assessing student learning by conducting oral tests and Quiz in the class.	Assessing the participation of students in small group discussions, brainstorming sessions etc. held in the class.
2.	Assessing students' learning by making them undertake project, field study etc.	Assessing skills and attitudes of students through their laboratory work and field investigations.
3.	Assessing student learning by observing his/her direct participation in the classroom activities.	Assessing students' learning by observing their involvement and interest in co-curricular activities.
4.	Using criterion referenced tests, assessing the level of learning achievement of students in terms of Grades.	Assessing student learning by comparing each student's learning achievement with the average performance of the class (Grade Norm); assessing student learning portfolio of each student.

2:03:3 Performance-based Assessment

Performance-based assessment is defined as evaluating students' ability to apply the skills and knowledge learned from a unit or units of study to complete a given task. Typically, the task challenges students to use their higher-order thinking skills to create a product or complete a process (Chun, 2010).

In the 1990s, performance-based assessments started becoming a valid alternative to traditional multiple choice tests. Now, more schools and colleges in the Western Countries are seeking authentic measures of student learning and performance-based assessments have become increasingly relevant.

Tasks can range in performance-based assessment, from a simple constructed response (e.g. short answer) to a complex design proposal of a sustainable neighbourhood. The most genuine assessment require students to complete a task that closely mirrors the responsibilities of a professional like artist, engineer, laboratory technician, financial analyst or consumer advocate.

Asking students to find why a particular law they have learned, can not be applied in a given situation or event will be testing their higher order thinking skills. Such test items find place in performance-based assessment. For example, asking students to tell why Boyle's law which states "At constant temperature, the pressure and volume of a given mass of gas are inversely proportional to each other" cannot be applied to, a man blowing a balloon, when both the volume and pressure of air inside the balloon increase. Similarly, asking students to perform the experiment to verify Boyle's law in the laboratory and record

the findings obtained, helps to assess students' performance.

2:03:3:01 Essential Components of a Performance-based Assessment

Test items or tasks given for assessing student's performance should consist one or more of the following constructs.

- i) Complex
- ii) Authentic
- iii) Process or Product oriented
- iv) Open-ended
- v) Time-bound

Normally, students are presented with an open-ended question that may produce several different correct answers. In the higher-level tasks, there is a sense of urgency for the product to be developed or the process to be determined, as in most real-world situations.

2:03:3:02 Teachers Preparing Students for Performance-based Assessments

Teaching-learning process that provides for 'Performance-based assessment' will have the following steps.

1. Identifying goals of the performance-based assessment

Teacher may fix the goal of assessing students' learning as evaluating his/her ability for critical thinking and problem-solving i.e. the teacher may try to evaluate students for not being dependent on others' and 'expressing individuality'. Here, the criteria for assessing student-performance is, he/she completing every step of a given task without seeking any help from other students or guidance from the teacher.

2. Selecting the appropriate course standards

After fixing the goals of assessment, the next step is selecting the 'Common Core Standard' to be addressed with this performance assessment. For example, if the XII Std. Maths teacher has taught a unit on 'Probability', then the assessment should measure students' **understanding** of 'Conditional probability' and rules of probability.

3. Reviewing assessment and identifying learning gaps

When students complete their learning in a particular content area of the subject, the teacher has to analyse the assignments submitted by the students to find what is missing in them. For example on the topic 'Probability', students' assignment papers or

worksheets may contain solving of numerical problems for the data given in the form of two-way frequency table. While assessing these assignment papers, the teacher may find a gap in students' learning in that the problems dealt with in those papers may be very little relevant to real world applications.

4. Designing the Scenario

In this step, after shooting of few brainstorming questions by the teacher, selecting the more appropriate ideas or answers presented by students, will take place. In our example, students may be asked to study the problems given in the text book involving two way frequency table and suggest an example for 'conditional probability' from our daily life, to each of the problems given. Preparing the question paper for an examination starts with fixing the content areas to be tested, identifying the instructional objectives and type of test items (i.e. different forms of items like essay type, short answer type, very short answer type and objective type) to be included and determining the weightages to be given for these three aspects (i.e. content areas, instructional objectives and type of test items) on the basis of the marks allotted out of the total marks of the question paper which leads to the development of '**Blue-Print**' of the question paper. Generally among the objective type test items, 'Fill in

the blanks', 'True/False', 'Multiple Choice' and 'Matching' type test items find place in the question paper.

According to the blueprint and keeping in mind, the instructional objectives to be tested and type of test items to be used, questions are to be developed from each content area and are assembled to form the question paper. 'Test items of the same form / type are grouped together. Brief directions for answering each type of test items are to be given. '**Scoring Scheme**' containing '**Cues**' for awarding marks (i.e. **Scoring cues**) for the responses of the examinees should be prepared well in advance, before start conducting the examination. This ensures objectivity and consistency in scoring the response sheets.

Examinees' answer scripts collected at the end of the examination, will be scored and awarded marks. Based on the marks obtained by examinees, they will be placed in the appropriate graded category of merit.

Instead of '**Teacher-made Tests**', 'Standardized Tests' could also be used for conducting examinations.

5. Gather / Create Suitable Learning Materials

In this step, students may be given an assignment to cite examples of practical applications in our daily

life for the different concepts learned in a particular content area of the subject.

6. Developing a Learning Plan

Teachers should develop the instructional plan, that not only helps students answer questions involved in performance assessment but also aims for making them understand the concepts in the lesson taught and develop their ability to fit them with practical applications in our day-to-day life. Teachers need to strike a balance between teaching the contents and preparing students for fitting the concepts learned with practical applications in our daily life i.e. preparing for the performance-based tasks.

2:03:4 Examination-based Assessment

The test that assess the subject knowledge and skills acquired by the students at the end of the instructional period is known as 'Examination'. The one who is subjected to the test is called 'Examinee'. Generally examination may be in the form of a written test, practical test or an on-line test. Tests used for evaluating the knowledge, skills, aptitude etc. of the examinees at the end of the instructional period, are alone called by the name 'Examination'.

2:03:4:01 Guidelines to Improve Examinations

1. Assessment should reflect the curriculum. Though it is not possible to assess everything in a curriculum, in an examination, the areas that are assessed should be the ones that are considered important.
2. Modes of assessment should reflect the curricular goals.
3. Test items that find place in an examination, besides measuring students' ability to recall and recognize the information learned, should also attempt to measure higher order learning outcomes like 'application', 'analysis', 'synthesis' etc.
4. Examinations should assess students' ability to apply their knowledge, not only in academic contexts but also in real life situations outside the school.
5. The content and form of items found in examination should be free from gender, ethnic group and location biases.
6. 60% of examination items should be of average difficulty level, 20% of items be easy and 20% of items be 'very difficult'.

7. As a blueprint is prepared before developing an examination question paper, preparing the 'scoring scheme' for the question paper is also equally important.
8. It is essential to ensure the 'content validity' of the question paper to be used in examination.
9. Care should be taken to see the question paper for the examination contains different types of test items.

2:04 Practices of Assessment of Students' Learning

The practices of assessment followed in the class for the formative evaluation of student learning will mostly be in the following patterns :

- i) Dialogue between the teacher and student.
- ii) Feedback through marking.
- iii) Peer and self-assessment

Let us discuss each one of these in detail.

2:04:1 Dialogue

Dialogue refers to sharing / exchanging ideas between two or more people.

Dialogue between the teacher and students in the class, could be used as a technique of assessment in the following ways :

- i) To assess the subject knowledge
- ii) To assess the learning outcomes
- iii) To assess the language ability of students
- iv) To assess and direct the learning activities of students
- v) To assess the learning achievement.

1. Assessing the Subject-knowledge

Before start teaching a topic, the teacher can find through dialogue, students' previous knowledge related to the topic to be taught. It becomes easy for the teacher to explain the concepts in the lesson or topic, using the previous knowledge of students.

2. Assessing the learning outcomes

At the terminal stage of teaching in the class, the teacher can assess through dialogue, how far the students have learned what are taught in the class, by asking oral questions. After completing a topic, the teacher can assess the learning achievement of students, through dialogue also by asking oral questions instead of employing written tests.

3. Assessing the language proficiency

Teachers could through dialogue assess the language ability of students. Some students in the

class may struggle to communicate what they know, using appropriate words; some may be fluent in expressing their ideas; some others may provide their responses in a word or two and that too when the teacher tries to extract the answer from their mouths.

4. Assessing the learning activities of students and putting them on the right track

Dialogue is useful for the teacher to evaluate different learning activities of students (like classroom assignment, home assignment etc.) and point out the corrections needed to improve.

'Interview' a different form of dialogue is useful for the teacher to find out the problems of children who find it difficult to learn and offer them remedial measures.

5. Assessing the learning achievement

Interview too could be used as a technique to assess learning achievement of a student. When this technique is used, the testee can neither provide vague replies without expressing his/her idea explicitly, nor copy other's responses. In a very short time, the depth of learning achievement of the testee could easily be assessed.

2:04:1:01 Strategies for Employing Dialogue

Dialogue could be employed in classroom teaching - learning as follows :

- i) Dialogue could be structured into many layers so that different dimensions of ideas could get expressed.
- ii) Getting engaged in flexible dialogue by relating many topics.
- iii) Getting engaged in dialogue that takes the form of question - answer.

By employing the above three techniques, dialogue could be made as the medium of classroom interactions with which the teacher could undertake the formative assessment of student - learning.

2:04:2 Feedback through Marking

The important aim of employing assessment tests in education is not only to measure student's learning achievement and express it in terms of marks but also providing feedback based on the findings obtained in tests, regarding the learning achievement expressed so as to improve it.

Providing feedback refers to making available the knowledge of results of one's own actions immediately.

It has been demonstrated by psychologists like **Skinner** and others that in the case of grown up adults, knowledge of results of one's own action itself serves as a positive reinforcer and enhances the level of performance just as praise or rewarding serves as an effective positive reinforcer in children to increase their level of performance. Thus 'feedback' is one of the means of achieving reinforcement of desired responses.

The teacher while scoring the student answers in an achievement test, should record his comments on the left margin of the answer script, against the answers mentioning which one requires corrections for improving clarity, which are to be modified, which ones are to be expanded and the information need to be added additionally, which answers are really good etc. While returning back the valued answer scripts to the students, the teacher should with the help of the comments he has recorded in each student's answer script, explain how the answers provided by him or her could be improved further.

For those students who have obtained below 40% of the total marks, it is enough to inform what are the correct answers for those against which comments are marked in their answer scripts. If more explanations are provided, such students may not be able to absorb

them. It is better for the teacher to ask such students to rewrite the correct answers once again.

For those students who score between 40% and 60% of total marks, the teacher can provide feed back in detail by pointing out the corrections to be made in their answers, explaining the aspects requiring refinement etc.

For those students scoring above 60% of the total marks, a detailed feedback could be provided by pointing out how their answers may be improved further supplementing with necessary illustrations, additional information needed, mentioning measurements with appropriate units etc.

2:04:2:01 Principles of Good Feedback Practices

1. Feedback provided should make students know what good performance is
2. Feedback provided by the teacher should help students to self-correct their responses
3. Opportunity is to be provided for students to resubmit their answers after improving them with the help of the feedback received
4. Followed by the returning of scored answer scripts, the teacher should arrange for a discussion with

students regarding the ways and means of improving their learning achievement

5. Enough time should be given for students to reflect on their learning achievement and the ways and means of improving it on the basis of the feedback received.
6. During the discussion in the feedback session, the teacher should make the students know the criteria for the appropriate answers expected by examiners.
7. Feedback provided should encourage positive motivational beliefs and self-esteem of students that leads to improvement in learning i.e. feedback should not involve severe criticism or condemning of students.

Stated briefly, providing feedback through marking should help students to know their learning achievement and its shortcomings, corrections and modifications required besides being helpful in motivating them to undertake measures of self-correction.

2:04:3 Peer and Self Assessment

2:04:3:01 Peer Assessment

Peer assessment is a process wherein students individually assess and grade each other's assignments or test papers based on predetermined list of

criteria. This practice helps to reduce the workload of teachers besides helping students to know the criteria for judging an assignment or answer as good. By evaluating other's answers / assignments, students are able to understand the different view points on various concepts or ideas contained in the subject contents. This helps to expand their subject knowledge and improve meta-cognitive skills like critically evaluating and self-reflecting on their own learning and skill development.

Peers should be provided with 'Guidelines for assessment' developed by the teacher, containing the list of criteria indicating the essential elements or aspects that are required to be present, in the answers or assignment. Then only every student can objectively evaluate the answer paper or assignment of other students without any bias. Further, it can not be assumed that all students know the correct answers. In due course of time, students could discuss among themselves and develop the criteria of assessment, based upon which undertake evaluation. The average of marks awarded by the members of the group is taken as the final mark or grade awarded for the student's assignment or answer paper.

In practice, in a crowded class with more number of students, it is not possible for the answer paper /

assignment of student getting evaluated by all other students in the class. Therefore in the normal classroom practice, the assignments / answer papers of students are distributed among themselves randomly so that each student can evaluate the learning achievement of another student in the class.

Based on the 'guidelines for assessment' provided by the teacher, the assignment / answer scripts of all students are evaluated simultaneously at the same time and marks / grades are awarded. One who evaluates the assignment / answer script will put his signature with name on the script evaluated. After evaluation is over, the answer script / assignment paper will be returned back to the respective students and asked to check whether evaluation is done properly according to the guidelines provided for assessment. The student complaining error in the assessment could show his / her assignment / answer script to the teacher and get the marks changed appropriately. If students are cautioned at the beginning of the assessment programme that the difference in marks reported in one's evaluation, will be subtracted from his / her own marks, then all students will undertake the assessment work honestly.

2:04:3:02 Self-Assessment

Self-assessment involves a student taking responsibility for monitoring and making judgements about the different aspects of his / her own learning, now and then. There are two stages in self-assessment of learning. They are :

- i) Determining the accepted criteria or standard for assessing the learning outcomes resulting from the learning of a particular area of the subject content.
- ii) Based upon the criteria, judging the extent of learning proficiency achieved.

Though self-assessment is a way of assessing the product of learning (product assessment), yet it is a learning process in itself. It is a way of improving student learning as it develops in students skills like 'knowing the assessment criteria' and evaluating learning achievement based on the accepted criteria. In fact, the term '**self-evaluation**' is more appropriate than the term 'self-assessment' as it develops the skills of judging the quality of one's own learning, identifying the shortcomings in learning and the ways and means of removing them. 'Self-assessment' leads to 'self-learning'. Hence it could be said that self-assessment is both summative and formative.

2:04:3:03 Uses of Peer and Self-Assessment

1. Students develop the skill of critically evaluating their own learning as well as that of others.
2. When students get involved in self or peer assessment, apart from the development of assessment skills, self-learning too takes place.
3. Self and peer assessment help students to discuss and decide the essential elements and aspects of quality work or good learning achievement.
4. Enable students to understand the subjective nature of judgements in assessments in many areas, which can only be substantiated by reference to other work i.e. self and peer assessment develop in students a better understanding of their subjectivity in judgement.
5. Understand the need for forming the criteria of assessment before starting evaluation of students' learning achievement.
6. Self and peer assessment reduce significantly teacher's work load and the time spent in marking students' answer scripts / assignments.
7. Self and peer assessment help to assess group work.

8. Self and peer assessment develop student involvement and responsibility.
9. Encourage students to reflect on their role and contribution to the process of group work.
10. Self and peer assessment help to provide feedback by combining learning achievement and assessment of learning.

Conclusion

In this unit, Evaluation of students' performance based on the concept, that education is reaching out what are to be taught to students i.e. evaluation of students based on transmission - reception model, which is otherwise known as 'behaviourist model' and its drawbacks, Changes that are taking place now in the assessment practices particularly the four important among them being (i) Use of constructivist approach in the assessment of students i.e. giving more importance for how students learn than what are learned (ii) Use of 'Continuous and Comprehensive Evaluation' (iii) Using grade symbols in the place of marks to denote learning achievement (iv) Adopting different techniques in evaluating multiple aspects of student development, Different kinds of assessment (practice based, evidence based, performance based and examination based), Some important practices of

assessment of student learning (dialogue, feedback through markings, peer and self-assessment) were discussed in detail.

Review Questions:

1. Explain the behaviourist's point of view of learning. (B) [Ans: 2:01]
2. Describe the 'Transmission-Reception' model of learning assessment based on the theory of 'Behaviourism'. Mention the drawbacks of this approach. (A) [Ans: 2:01 + 2:01:1 + 2:01:1:01]
3. Mention the drawbacks of learning assessment based on the 'Transmission-reception' model. (C) [Ans: 2:01:1:01]
4. Write a note on changes that are taking place now in the assessment practices. (B) [Ans: 2:02]
5. Explain the constructivist's approach in the assessment of student learning, stating the tools and methods employed. (A) [Ans: 2:02:1:01 + 2:02:1:02 + 2:02:1:03]
6. Describe the role and functions of the teacher in the assessment of student learning in a constructivist classroom. (B) [Ans: 2:02:1:04]

7. Explain the 'Continuous and Comprehensive Evaluation Scheme' in the assessment of student learning. (A) [Ans: 2:02:2:01 to the end of 2:02:2:04]
8. State the merits and demerits of 'Continuous and Comprehensive Evaluation Scheme'. (B) [Ans: 2:02:2:05 + 2:02:2:06]
9. What is the role and functions of the teacher in the 'Continuous and Comprehensive Evaluation Scheme' (C) [Ans: 2:02:7]
10. Describe the following techniques of evaluation in assessing the multiple aspects of student development : (i) Project (ii) Seminar (A) [Ans: In 2:02:3:01, those under (A), (C) and (F) + 2:02:3:02 + (A), (B) and (E) of 2:02:3:02]
11. Explain the 'project-based learning', stating its merits and demerits. (A) [Ans: (A), (B), (C), (D) and (E) of 2:02:3:01]
12. What is meant by a seminar? Explain the seminar method of teaching, stating its merits and demerits. (A) [Ans: A, B, C and D of 2:02:3:02]
13. Explain the concept of 'assignment' and how it could be used as a teaching device. State the advantages and disadvantages of this method. (A) [Ans: A, B, C and D of 2:02:3:03]

14. State the ways and means of improving assignment as a teaching device. (B) [Ans: (E) of 2:02:3:03]
15. Describe the method of assessing student performance based on assignments submitted. (A) [Ans: (A), (B) and (F) of 2:02:3:03]
16. Explain the meaning of 'Student learning portfolio' and give a suitable definition for the same. (B) [Ans: 2:02:4:01 + 2:02:4:02]
17. Explaining the concept of 'Student learning portfolio' describe how it could be used to assess student-learning. (A) [Ans: 2:02:4:01 + 2:02:4:02 + 2:02:4:03]
18. Explain how portfolio assessment of student-learning is designed. (A) [Ans: 2:02:4:02 + 2:02:4:04]
19. State the advantages and limitations of portfolio assessment of student learning. (B) [Ans: 2:02:4:05 + 2:02:4:06]
20. What do you mean by grading student performance? Explain the main features of the grading scheme followed by the Tamilnadu Government. (A) [Ans: 2:02:5 + 2:02:5:01]
21. What are the uses of grading system in the place of marks in the assessment of student performance? (C) [Ans: 2:02:5:02]

22. Explain the concept of 'Practice-based assessment' and how it could be implemented in the classroom situation. (A) [Ans. 2:03:1]
23. Explain the concept of 'Evidence-based assessment' and the method of implementing it. (A) [Ans. 2:03:2]
24. Explain the concept of 'Performance-based assessment' and the method of implementing it. (A) [Ans. 2:03:3]
25. Explain the concept of 'Examination-based assessment'. (B) [Ans. 2:03:4]
26. Suggest the ways and means to improve 'Examinations' at the school level. (B) [Ans. 2:03:4:01]
27. Mention the assessment practices followed in the evaluation of student-learning. (C) [Ans. 2:04]
28. How classroom dialogue could be used to evaluate student learning? (B) [Ans. 2:04:1]
29. Explain the concept of feedback through marking? (B) [Ans. 2:04:2]
30. State the principles of good feedback practices. (B) [Ans. 2:04:2:01]